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Cloud-Edge Retrieval-Augmented Agent Tutor for Trustworthy Personalized Learning in Resource-Constrained Higher Education

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***Abstract*—Generative AI tutors are attractive for universities that lack enough human tutoring capacity, but unbounded cloud chatbots are difficult to audit, costly to operate, and risky when advice is not grounded. This paper presents a cloud-edge retrieval-augmented agent tutor for trustworthy personalized learning in resource-constrained higher education. The workflow combines early-risk learning analytics, local evidence retrieval, policy checking, route-aware cloud escalation, and an explicit run record. On the Open University Learning Analytics Dataset (OULAD), the executed pipeline processed 32,593 student-course records, produced 8,149 test predictions, audited 160 high-risk feedback cases, and generated a full-test routing audit. Logistic regression achieved Macro-F1=0.757, AUC=0.838, and at-risk recall=0.693 from first-28-day activity; across five split seeds, Macro-F1 was 0.757±0.002. Feature ablation, query ablation, and route-policy stress tests identify which design choices matter. The contribution is a reproducible AIoTC-style architecture and evidence package, bounded to offline public-data evaluation rather than classroom learning gains or hardware latency benchmarking.**

***Index Terms*—retrieval-augmented generation, LLM agents, learning analytics, edge AI, cloud computing, trustworthy AI, personalized learning, SDG 4**

I. Introduction

Universities are rapidly testing generative AI for tutoring and feedback, including large-scale automated feedback for language learning [1], [2]. The education literature is also cautious about hallucination, academic-integrity, transparency, and governance risks [1]. These risks are sharper in resource-constrained institutions. A useful AI tutor must personalize support, but it cannot send every learner interaction to a cloud model or issue advice that cannot be traced to evidence.

Three technical streams are relevant but often remain separate. Learning analytics can identify learners who may need early support [3]–[5]. Retrieval-augmented generation (RAG) can ground generated language in external evidence, although retrieval does not remove the need for faithfulness evaluation [6]–[8]. LLM-agent work provides a way to separate sensing, tool use, policy checking, and response composition [9]. Edge intelligence and mobile-edge LLM research further motivate routing some computation away from the cloud when latency, privacy, bandwidth, or cost matters [10], [11].

This paper joins these streams into a compact AIoTC education service. The technical novelty is a logged control loop linking risk scoring, course-context retrieval, evidence-marker auditing, policy blocking, and cloud-edge routing for one learner record. This differs from educational RAG that mainly grounds answers and learning analytics that mainly predicts risk. This positioning is consistent with recent work on auditability in AI-integrating learning-analytics systems [12]. We contribute (i) a route-aware RAG agent architecture, (ii) a reproducible OULAD experiment with prediction, ablation, retrieval, agent, and routing artifacts, and (iii) an explicit claim boundary that avoids unperformed classroom, human-rating, or edge-device claims.

II. Method

Fig. 1 shows the system. The edge side computes early-risk scores and local retrieval over course statistics and support-policy passages. The agent executes a fixed trace: sense risk, retrieve evidence, check policy, compose feedback, and route delivery. Sense risk reads the score and early activity; retrieve

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evidence selects course/policy passages; check policy blocks unsupported or sensitive claims; compose feedback emits evidence-marked guidance. A high-risk record with few first-28-day clicks and low early assessment mean is routed to cloud handling when retrieval confidence is low, after citing course evidence and support policy.

For each student-course record i, early features $x_i^{0:28}$ are computed from the first 28 days of virtual learning environment (VLE) activity, registration data, assessment submissions, and course/learner descriptors. The binary target in Eq. (1) is

$$y_i = \mathbf{1}\{\text{final}_i \in \{\text{Fail}, \text{Withdrawn}\}\}. \tag{1}$$

The edge predictor in Eq. (2) estimates an at-risk probability by a regularized logistic objective:

$$\begin{aligned} r_i &= \sigma(w^\top z_i + b), \\ \mathcal{L} &= -\sum_i \{y_i \log r_i + (1-y_i)\log(1-r_i)\} + \lambda \|w\|_2^2, \end{aligned} \tag{2}$$

where z_i is the encoded feature vector. We compare this model with random forest and Extra Trees under the same split. Demographic fields are used for subgroup auditing and reproducibility, not for stereotype-based feedback.

For feedback, a query q_i combines course code, presentation, risk score, early clicks, active days, and early assessment mean. A TF-IDF retriever ranks evidence passages by cosine similarity as Eq. (3):

$$\mathcal{R}_k(q_i) = \text{TopK}_{d \in \mathcal{D}} \cos(\phi(q_i), \phi(d)). \tag{3}$$

Retrieval confidence is $c_i = \max_{d \in \mathcal{R}_k(q_i)} \cos(\phi(q_i), \phi(d))$. The audit uses four claim slots: risk status, behavioral evidence, course comparison, and action. A supported claim must cite [E1] or [P1], match course context, and avoid protected or sensitive inferences. For M expected claim slots and m_i observed evidence markers, unsupported-claim rate is

$$u_i = 1 - \min(m_i/M, 1). \tag{4}$$

Using Eq. (4) as the grounding audit, the route in Eq. (5) is assigned by a deterministic policy with confidence threshold $\tau_c = 0.04$:

$$\rho_i = \begin{cases} \text{edge-only}, & r_i < 0.55,\ c_i \geq \tau_c, \\ \text{edge-first}, & 0.55 \leq r_i < 0.75,\ c_i \geq \tau_c, \\ \text{cloud}, & r_i \geq 0.75 \text{ or } c_i < \tau_c. \end{cases} \tag{5}$$

Algorithmically, each record follows six steps: construct z_i, score r_i with Eq. (2), rank $\mathcal{R}_k(q_i)$ with Eq. (3), compose evidence-marked feedback, audit u_i, and export the route ρ_i from Eq. (5). Latency and token values are route-level proxies, not hardware measurements.

III. Experiment and Run Record

The experiment uses OULAD [3], a public dataset containing course metadata, assessments, VLE resources, student information, registrations, assessment submissions, and VLE clicks. The pipeline builds a 32,593-row first-28-day feature table with an at-risk rate of 0.528 and uses a stratified 75/25 split with seed 2026. The test set contains 8,149 records.

TABLE I
Executed pipeline record and extracted artifacts.

Item	Evidence generated in this project
Command	`python3 03_Experiments/run_pipeline.py`
Environment	Python 3.9.6; scikit-learn 1.6.1; NumPy 2.0.2
Data manifest	7 OULAD CSV files with byte sizes and SHA-256 hashes
Feature/prediction outputs	32,593 feature rows; 8,149 test-prediction rows
Feedback/routing outputs	160 feedback-audit rows; 8,149 route-audit rows
Ablation/stress outputs	5 split seeds; 3 time windows; 4 feature variants; 3 query variants; 4 route policies
Claim boundary	Offline public-data experiment; no classroom or hardware claim

Table I summarizes the run record. In addition to metric summaries, the project stores the processed feature table, test predictions, feedback-audit CSV, agent-audit CSV, route-audit CSV, sustainability indicators, run log, and command transcript. This matters because the paper's claim is not that an abstract framework "should work," but that an executable pipeline produced traceable artifacts.

TABLE II
First-28-day at-risk prediction on OULAD.

Model	Acc.	Macro-F1	AUC	Risk Rec.
Logistic Reg.	0.757	0.757	0.838	0.693
Random Forest	0.756	0.756	0.842	0.668
Extra Trees	0.757	0.756	0.842	0.648

Table II shows that logistic regression gives the best Macro-F1. The tree ensembles provide close AUC but lower at-risk recall. The selected model was rerun over five stratified split seeds, giving Macro-F1 of 0.757±0.002, AUC of 0.841±0.004, and at-risk recall of 0.694±0.007. The largest measured recall gap among gender, disability, and low-IMD groupings is 0.118, so the system records a fairness warning rather than claiming fairness certification.

Fig. 2 summarizes the evidence matrix. Window sensitivity showed that Macro-F1 increased from 0.700 at day 7 to 0.725 at day 14 and 0.757 at day 28, while AUC increased from 0.768 to 0.838. Feature ablation showed that removing VLE activity reduced Macro-F1 to 0.727 and removing assessment features reduced it to 0.732; an administrative/profile-only variant fell to 0.638. These results support the design choice of using early behavioral and assessment signals together.

For RAG evaluation, we selected the 160 highest-risk test records. Retrieval reached precision@1=1.00 and hit@3=1.00 for the exact course-presentation statistics document. A query ablation exposed the boundary: removing course identifiers reduced precision@1 and hit@3 to 0.006. Grounded feedback reached evidence coverage of 1.00 and unsupported-claim rate of 0.00; generic advice reached evidence coverage of 0.00 and unsupported-claim rate of 1.00. The local retrieval

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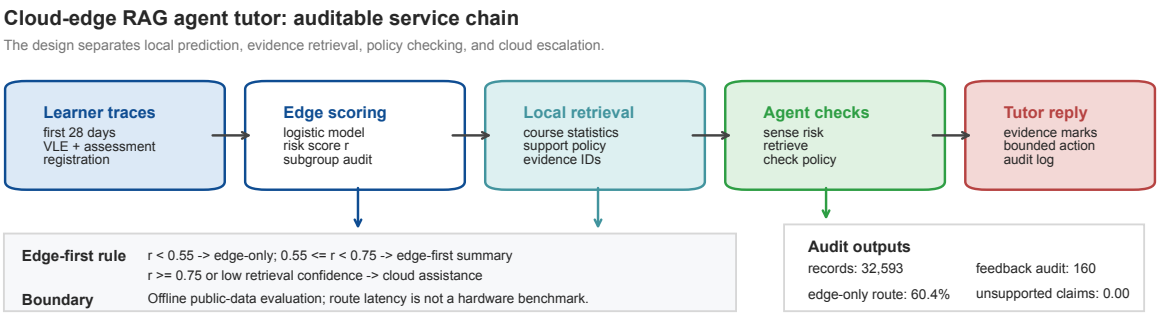


Fig. 1. Auditable cloud-edge RAG tutor architecture. The workflow separates learner traces, edge scoring, local retrieval, agent policy checks, route selection, and logged feedback so that each personalized recommendation can be traced to evidence and a delivery route.

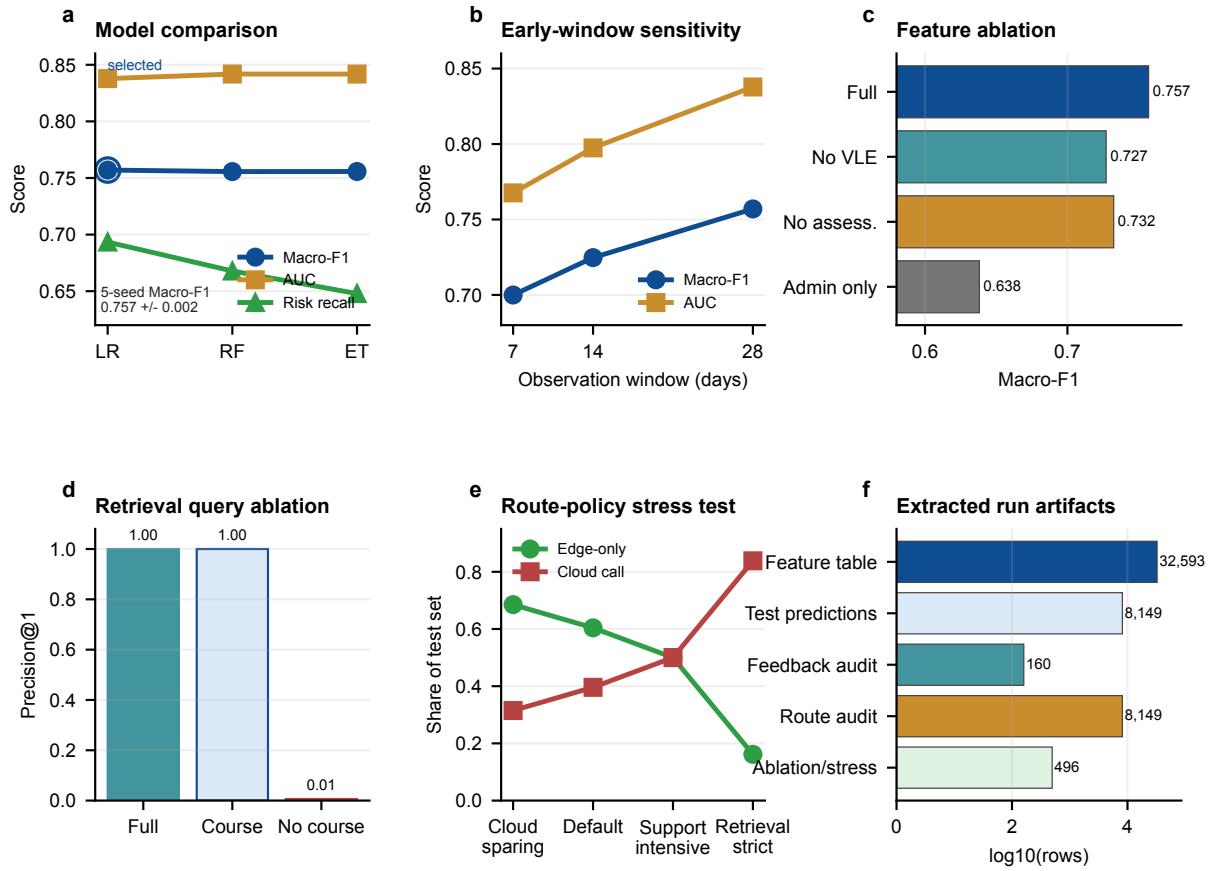


Fig. 2. Experimental evidence matrix. (a) Three compact models were trained and compared on the same split; the selected model was also checked across five split seeds. (b) Longer early-observation windows increased Macro-F1 and AUC. (c) Feature ablations show that removing VLE or assessment signals reduces balanced performance. (d) Retrieval fails when course context is removed from the query. (e) Route-policy stress tests expose the resource-support trade-off. (f) The executed pipeline produced row-level artifacts and ablation/stress-test records.

measurement in the latest run was about 0.24 ms per feedback item.

The routing audit was run on the full 8,149-record test set. The default policy assigned 60.4% of records to edge-only handling, 13.7% to edge-first cloud summary, and 25.9% to cloud-assisted handling. Route-policy stress tests changed cloud-call share from 31.5% under a cloud-sparing policy to 50.0% under a support-intensive policy and 83.8% under a retrieval-strict policy. These values indicate a controllable resource-support trade-off; they should not be read as device benchmarks.

IV. Discussion

The results support the design choice of treating learner-facing AI as an auditable service. Learning analytics supplies

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a support trigger, RAG supplies evidence grounding, the agent trace supplies governance, and route-aware delivery supplies resource control, consistent with calls for ethical educational AI [13].

The strongest evidence is reproducibility: the same script regenerates the feature table, model metrics, seed checks, ablations, feedback audit, route audit, and figures from public data. The run transcript records the command, software versions, row counts, scores, audit rows, and figure outputs, making the architecture checkable rather than conceptual.

The evidence is intentionally conservative. The selected model is not claimed to be best in all settings; it is selected because it gives the best Macro-F1 in this run and the highest at-risk recall among tested compact baselines. The feedback audit checks grounding, prohibited inferences, and template actionability, but not free-form conversation, learner satisfaction, or teaching effectiveness.

The main deployment boundary is cloud-edge measurement: the route audit covers every test record, but latency and token values remain proxies. Deployment still requires edge-device profiling, network measurements, human ratings, and institutional review before classroom-impact claims.

For AIoTC education systems, this boundary is useful. The workflow turns a tutor into inspectable decisions: which learners were flagged, which evidence passages were retrieved, which claims were allowed, and which route handled the request. The recall-gap warning is a deployment gate: administrators review subgroup recall before launch, adjust thresholds or add human review when gaps persist, and never use the flag for punitive automation. Prediction remains a support trigger rather than an automated intervention. Route thresholds can change without retraining the predictor, helping resource-constrained universities audit the service before live teaching.

V. Conclusion

This paper presented a cloud-edge RAG agent tutor for trustworthy personalized learning in resource-constrained higher education. In a reproducible OULAD experiment, risk analytics, evidence-grounded feedback, and edge-first routing produced an inspectable AIoTC workflow under a conservative claim boundary. Future deployment work should test classroom learning gains and hardware latency.

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